# Lucas Flowers

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## Summary

Robotics Engineering Technology student in Purdue's John Martinson Honors College who founded and grew an engineering organization to 10+ members and $20K in sponsorship funding, building real-time control systems and integrating sensors, actuators, and electronics on competition robots. Hands-on fabrication experience in a community manufacturing makerspace, with proficiency in Java, Python, C, CAD, and rapid prototyping.

## Education

**Purdue University, John Martinson Honors College** — West Lafayette, IN
*Bachelor of Science in Robotics Engineering Technology* — *Expected May 2030*

**Phoenix Union Bioscience High School** — Phoenix, AZ
*Engineering-Focused Curriculum* — *May 2026*

## Technical Skills & Certifications

**Programming:** Java, Python, C, JavaScript, SQL, HTML/CSS
**Robotics & Controls:** Control systems, autonomous systems, embedded systems, sensor integration, actuator control, real-time systems, subsystem integration
**Manufacturing & Fabrication:** Prototyping, rapid prototyping, electronics, electrical wiring, mechanical assembly, 3D printing, welding, metalworking
**Software & Tools:** CAD, Autodesk Fusion 360, WPILib (robotics control-system library), Git, GitHub, Linux, VS Code
**Certifications:** Autodesk Fusion Certified User; IT Specialist Certifications (Pearson) in Java, Python, JavaScript, and HTML & CSS; Arizona Technical Skills Assessments in Software & App Design and Engineering

## Engineering Projects

**Makers of Change — Southwest Human Development** — Phoenix, AZ
*Embedded Systems Engineering Finalist* — *Oct. 2025*

- Selected as one of nine statewide finalist teams from 90+ teams across Arizona
- Engineered an embedded control system for an assistive-technology prototype improving accessibility for a child with disabilities
- Integrated hardware and control logic on a five-student team, iterating the prototype through testing and design revisions based on user feedback

**High-Altitude Weather Balloon Camera System** — Phoenix, AZ
*Project for Bioscience High School E-TECH Club* — *Feb. 2025*

- Built a Raspberry Pi dual-camera imaging system with a Flask web interface for in-flight monitoring
- Launched the payload to 90,000 ft, capturing two hours of 4K footage, validating the system's design for near-space conditions

## Engineering Experience & Leadership

**Phoenix Forge Makerspace — GateWay Community College** — Phoenix, AZ
*Engineering Intern* — *Sept. 2025 – May 2026*

- Fabricated engineering prototypes in a community manufacturing environment using metalworking, welding, woodworking, 3D printing, and rapid-prototyping workflows
- Applied technical problem solving, measurement, and diagnostics while following safe shop practices on fabrication equipment
- Guided makerspace members through equipment operation, fabrication, and troubleshooting

**FIRST Robotics Competition — Team Epsilon** — Phoenix, AZ
*Founder, President, Lead Programmer & Control Systems Lead* — *May 2025 – May 2026*

- Founded and scaled competition robotics team to 10+ students, raising $20K in sponsorship funding
- Architected a vision-based auto-aim/shoot control system using fiducial marker detection, integrating sensors, actuators, and real-time autonomous and driver-operated control on competition robots
- Directed control systems across multiple robots and mentored teammates in programming, electrical integration, and CAD
- Earned the team's Rookie All Star Award at the Arizona Robotics League off-season competition (Summer 2025) and at the FIRST Utah Regional for engineering design and community impact

**Wentworth Institute of Technology — Impact Lab** — Boston, MA
*Participant, Robotics Engineering Pre-College Program* — *Jul 2024*

- Completed college-level robotics coursework exploring kinematics, automation, sensors, and ethical engineering design

## Organizations

**Purdue Professional Ambassadors** — Aug. 2026 – Present
**Purdue Orbital** — Avionics Systems — Aug. 2026 – Present
**Purdue Electric Racing** — Low Voltage Systems — Aug. 2026 – Present